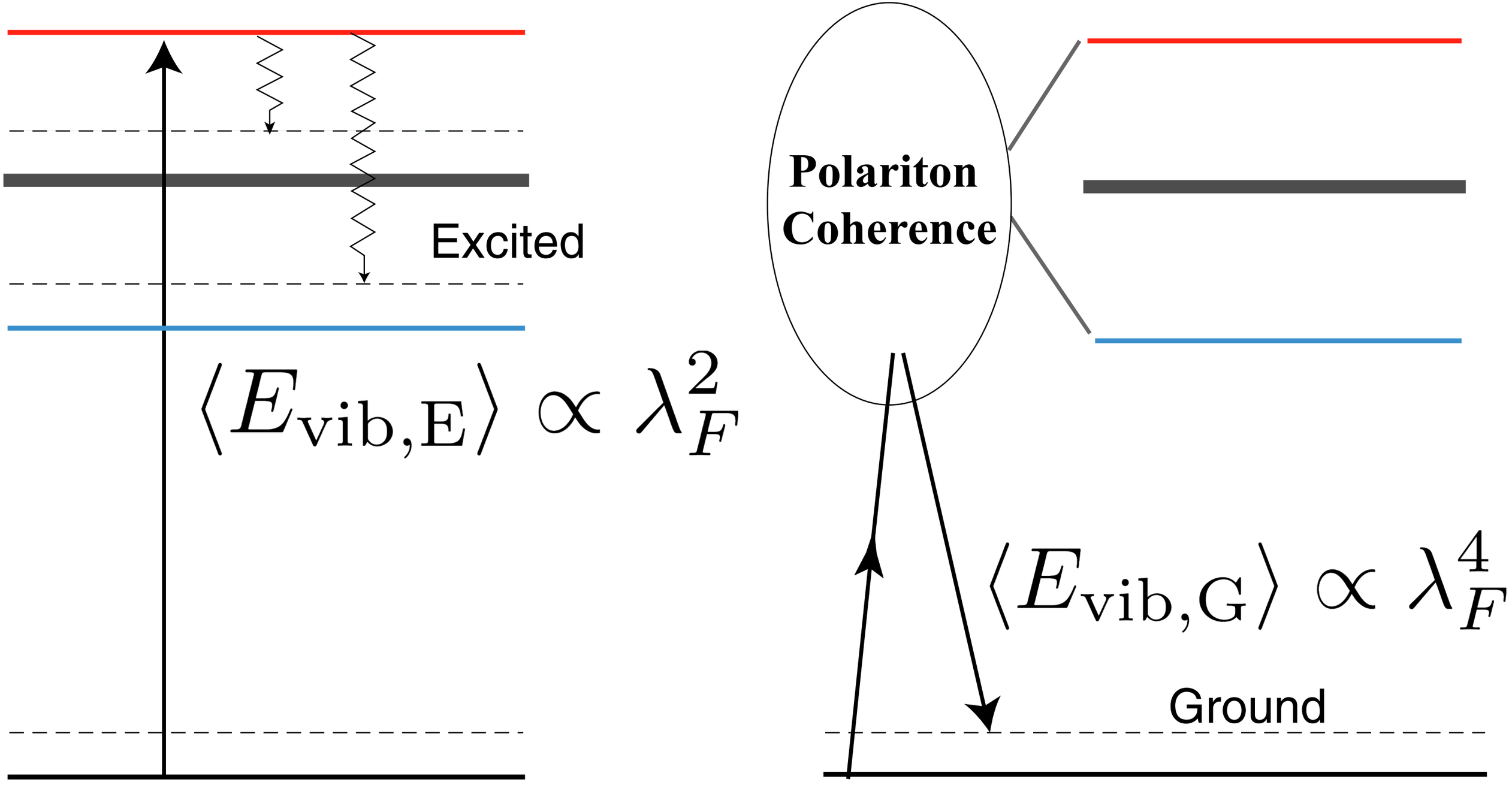
Excited
$\langle E_{\mathrm{vib,E}} \rangle \propto \lambda_F^2$
Polariton
Coherence
$\langle E_{\mathrm{vib,G}} \rangle \propto \lambda_F^4$
Ground